KING KHALID UNIVERSITY

COLLEGE OF COMPUTER SCEINCE

ACCESSORY STORE

By

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ABSTRACT

The **Accessory Store Project** is a comprehensive platform that covers all stages of jewelry manufacturing — starting from the extraction of gold or silver, through the processes of production and design, up to the secure and professional delivery of the final piece to the customer. The project aims to provide an integrated online shopping experience that enables users to browse accessories, select their favorite products, and purchase them easily, while ensuring efficient inventory management and order tracking.

The project applies **software engineering principles** to develop a structured and scalable platform that follows the **Software Development Life Cycle (SDLC)**, including requirement gathering, system analysis, design modeling, planning, and implementation. It enhances the user experience, simplifies sales and purchasing operations, and improves inventory management efficiency, making it a practical example of how technological solutions can be applied in the accessory trade sector.

Additionally, the project contributes to supporting the **digital transformation** of small and medium-sized accessory stores by providing them with an effective electronic tool for marketing and sales management. It also serves as a **training opportunity** to apply theoretical software engineering concepts in a real-world environment, including skills such as analysis, design, documentation, and teamwork.

Furthermore, the project aims to develop a prototype that can later be expanded into a fully functional system supporting multiple languages and offering an enhanced interactive user interface. It also emphasizes **data security and transaction safety**, making it a project that effectively combines both technical and practical aspects to serve the commercial and operational domains simultaneously.

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CHAPTER 1: INTRODUCTION

Problem Statement

Many small accessory shops face difficulties in reaching a wider audience and managing their sales efficiently. Customers also struggle to find specific products that combine high quality, attractive appearance, and reasonable price. The Accessory Store system addresses these issues by creating an online platform that allows customers to browse, search, and purchase accessories easily, while enabling the store to manage inventory, track sales, process customer orders efficiently, and market products in an appealing and effective way.

Objectives

- Designing an accessories website that suits the target audience.
- Enabling customers to view product details and prices easily and quickly.
- Providing secure options for user login and payment.
- Managing customer data securely.
- Ensuring complete customer satisfaction and gaining repeat customers.

Scope

The scope of this project covers the design and documentation of an online Accessory Store. It includes user registration, browsing of available items, order placement, and payment processing. Administrative functionalities such as inventory updates, user management, and sales reports are also part of the scope. The project will focus on the design phase and not on physical implementation or deployment.

Significance

This project holds significant importance in integrating the art of accessory making with advanced e-commerce, providing stores with the opportunity to reach a larger audience. It also offers an opportunity to apply software engineering principles to solve real-world problems faced by many stores.

Assumptions & Constraints

Assumptions:

- Users have access to smartphones or computers to browse the website easily.
- There is a stable internet connection to support smooth browsing and transactions.
- The online payment system supports both local and international currencies.
- Product images and prices are accurately provided and regularly updated by suppliers.
- The development team has the necessary tools and software to design the website and database.
- Customers are familiar with using online shopping platforms to complete purchases.

Constraints:

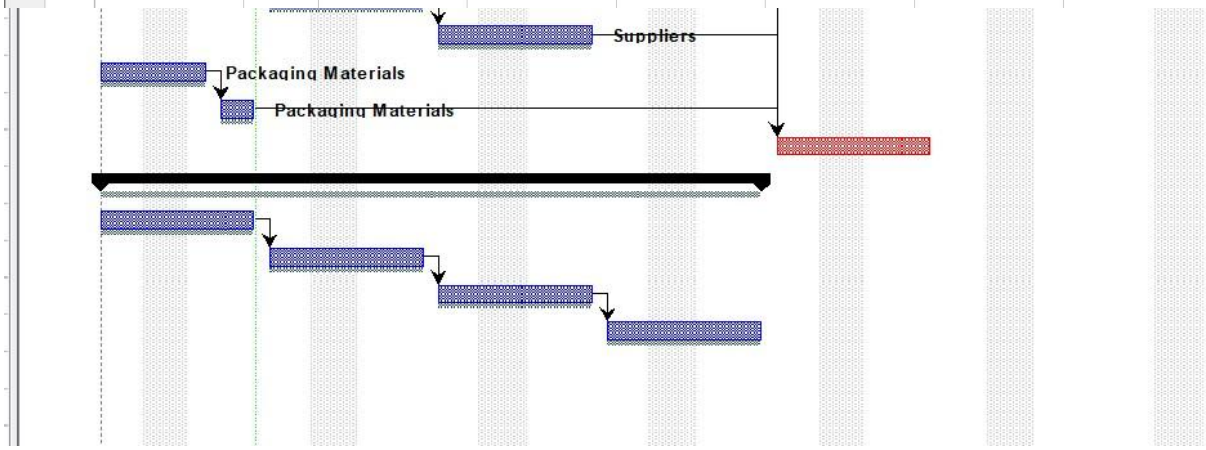
- The project budget is limited, which may affect the inclusion of additional features.
- The project timeline is short, restricting delivery to design and documentation only.
- The project depends on product data and images provided by suppliers, which may be delayed.
- The development must use predefined tools and technologies approved for the project.

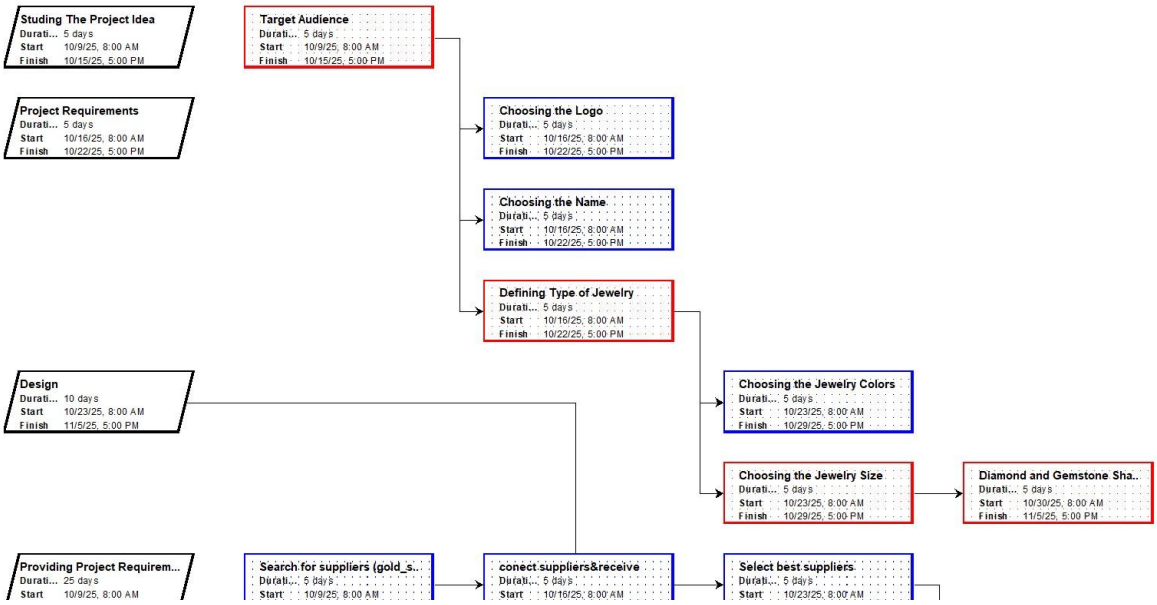
CHAPTER 2: PROJECT PLAN

Gantt Chart

The following Gantt chart presents a detailed schedule of the accessories project, including the resources assigned and the total duration required to complete the entire project

		Name	Duration	Start	Baseline Cost	Baseline Finish	Finish	Predecessors	Resource Names
1		**Studing The Project Idea**	5 days	**10/9/25, 8:00 AM**	SAR0.00	10/16/25, 5:00 PM	**10/15/25, 5:00 PM**		
2		Target Audience	5 days	10/9/25, 8:00 AM	SAR0.00	10/16/25, 5:00 PM	10/15/25, 5:00 PM		
3		**Project Requirements**	5 days	**10/16/25, 8:00 AM**	SAR26200.00	11/3/25, 9:00 AM	**10/22/25, 5:00 PM**		
4		Choosing the Logo	5 days	10/16/25, 8:00 AM	SAR24500.00	10/27/25, 9:00 AM	10/22/25, 5:00 PM	2	Designer
5		Choosing the Name	5 days	10/16/25, 8:00 AM	SAR0.00	11/3/25, 9:00 AM	10/22/25, 5:00 PM	2	
6		Defining Type of Jewelry	5 days	10/16/25, 8:00 AM	SAR1700.00	10/24/25, 9:00 AM	10/22/25, 5:00 PM	2	Gold;Silver
7		**Design**	10 days	**10/23/25, 8:00 AM**	SAR46700.00	12/23/25, 11:00 AM	**11/5/25, 5:00 PM**		
8		Choosing the Jewelry Colors	5 days	10/23/25, 8:00 AM	SAR19350.00	11/17/25, 10:00 AM	10/29/25, 5:00 PM	6	Jewelry designer
9		Choosing the Jewelry Size	5 days	10/23/25, 8:00 AM	SAR25350.00	11/24/25, 10:00 AM	10/29/25, 5:00 PM	6	Jewelry designer
10		Diamond and Gemstone Sha	5 days	10/30/25, 8:00 AM	SAR2000.00	12/23/25, 11:00 AM	11/5/25, 5:00 PM	9	Jewelry Design Tools
11		**Providing Project Requiren**	25 days	**10/9/25, 8:00 AM**	SAR12300.00	12/30/25, 11:00 AM	**11/12/25, 5:00 PM**		
12		Search for suppliers (gold_s	5 days	10/9/25, 8:00 AM	SAR4000.00	10/15/25, 5:00 PM	10/15/25, 5:00 PM		Suppliers
13		conect suppliers&receive	5 days	10/16/25, 8:00 AM	SAR4000.00	10/22/25, 5:00 PM	10/22/25, 5:00 PM	12	Suppliers
14		Select best suppliers	5 days	10/23/25, 8:00 AM	SAR4000.00	10/29/25, 5:00 PM	10/29/25, 5:00 PM	13	Suppliers
15		Prepare packaging materials	3 days	10/9/25, 8:00 AM	SAR150.00	10/13/25, 5:00 PM	10/13/25, 5:00 PM		Packaging Materials
16		Finalize packaging design &	2 days	10/14/25, 8:00 AM	SAR150.00	10/15/25, 5:00 PM	10/15/25, 5:00 PM	15	Packaging Materials
17		Order initial quantities of ma	5 days	11/6/25, 8:00 AM	SAR0.00	12/30/25, 11:00 AM	11/12/25, 5:00 PM	7;14;16	
18		**Shipping Companies&Exter**	20 days	**10/9/25, 8:00 AM**	SAR0.00	11/5/25, 5:00 PM	**11/5/25, 5:00 PM**		
19		Research local and internati	5 days	10/9/25, 8:00 AM	SAR0.00	10/15/25, 5:00 PM	10/15/25, 5:00 PM		
20		Contact companies & get pr	5 days	10/16/25, 8:00 AM	SAR0.00	10/22/25, 5:00 PM	10/22/25, 5:00 PM	19	
21		Compare services and selec	5 days	10/23/25, 8:00 AM	SAR0.00	10/29/25, 5:00 PM	10/29/25, 5:00 PM	20	
22		Sign initial agreement & run	5 days	10/30/25, 8:00 AM	SAR0.00	11/5/25, 5:00 PM	11/5/25, 5:00 PM	21	





		Name	RBS	Type	E-mail Address	Material Label	Initials	Group	Max. Units	Standard Rate	Overtim
1		Gold		Material	Material		G	G		SAR1000.00	
2		Silver		Material	Material		S	S		SAR700.00	
3		Jewelry Design Tools		Material	Material		J	J		SAR2000.00	
4		Suppliers		Work	Work		S	S	400%	SAR100.00/hour	
5		Packaging Materials		Material	Material		P	P		SAR150.00	
6		Jewelry designer		Work	Work		J	J	200%	SAR150.00/hour	S
7		Designer		Work	Work		D	D	100%	SAR500.00/hour	

Project Timeline Description

The Accessory Store project begins with requirement analysis, followed by system design, documentation, and presentation. The overall duration spans approximately 14 weeks, starting from project initiation to final presentation. Initial weeks focus on defining the problem and gathering requirements, followed by UML modeling, diagram design, and report preparation in later weeks. Milestones are defined for each phase to ensure steady progress.

Task Allocation

Week	Team Member	Role	Task
1-2, 10-12, 14	Rawan Muflih	Project Coordinator	• Understanding the project concept, identifying goals and targeting users. ' • Preparing the initial project schedule and high-level Gantt Chart. • Review work progress and ensure adherence to the schedule. ' • Reviewing full documentation, ensuring consistency, and preparing team for final presentation.
2-3, 8-10, 12-13	Reem Ali	Documentation Specialist	• Write the initial project report (Chapters 1 & 2). • Document functional and non-functional requirements. • Ensure consistent structure and writing style. •Write Chapters 3 & 4 (Requirements & System Design). • Ensure clarity, formatting, and technical accuracy. • Write the mid and final chapters (Conclusion, Future Work). • Format the full report and finalize the document. • Prepare the presentation slides.

3-4, 5-7, 14	Waad Abdulaziz	System Designer	• Create UML diagrams (Class & Activity Diagrams). • Design product features (colors, sizes, shapes, attributes). • Analyze entity relationships within the system. • Review and validate the system design •Enhance the overall system architecture and consistency. •Review all design diagrams
4-5, 6-8, 14	Layan Mohammed	System Analyst	• Create the Use Case Diagram to illustrate user system interaction. • Define all system functions and use case scenarios. • Review and analyze use cases to ensure completeness. • Collaborate with the designer to align use cases with workflow diagrams. • Review analysis chapters.

CHAPTER 3: REQUIREMENTS SPECIFICATION

3.1 Functional Requirements

Functional requirements define the specific behaviors and functions the system must perform. These are based on the system's scope, which includes user registration, browsing, ordering, payment, and administrative tasks.

ID	Requirement Description	Priority
User Management		
FR-001	The system shall allow new users (customers) to register for an account.	High
FR-002	The system shall provide secure login options for registered users.	High
FR-003	The system shall allow administrators to manage user accounts.	Medium
FR-004	The system shall securely manage all customer data.	High
Product & Browsing		
FR-005	The system shall enable customers to browse the list of available accessory items.	High
FR-006	The system shall allow customers to view product details and prices easily and quickly.	High
FR-007	The system shall provide a search function for customers to find specific products.	High
E-Commerce		
FR-008	The system shall enable customers to select their favorite products for purchase.	High
FR-009	The system shall allow customers to place an order.	High
FR-010	The system shall provide secure payment processing options.	High
Administrative Functions		
FR-011	The system shall allow administrators to manage and update inventory.	High

FR-012	The system shall allow administrators to process customer orders efficiently.	High
FR-013	The system shall allow the store to track sales.	Medium
FR-014	The system shall allow administrators to generate sales reports.	Medium
FR-015	The system shall enable the store to market products in an appealing way.	Medium

3.2 Non-functional Requirements

Non-functional requirements describe the quality attributes and constraints of the system.

- **Usability:** The system must be easy to use, allowing customers to browse, search, and purchase accessories without difficulty. The website design must be suitable for the target audience and provide an appealing, interactive user interface.
- **Performance:** The system must respond quickly, especially when customers are viewing product details and prices.
- **Security:** The system must prioritize data security and transaction safety. This includes providing secure options for user login, secure payment processing, and ensuring customer data is managed securely.
- **Scalability:** The platform must be developed as a structured and scalable system. The initial prototype is intended to be expanded into a fully functional system in the future.
- **Maintainability:** Administrators must be able to maintain updated product data.

CHAPTER 4: SYSTEM DESIGN

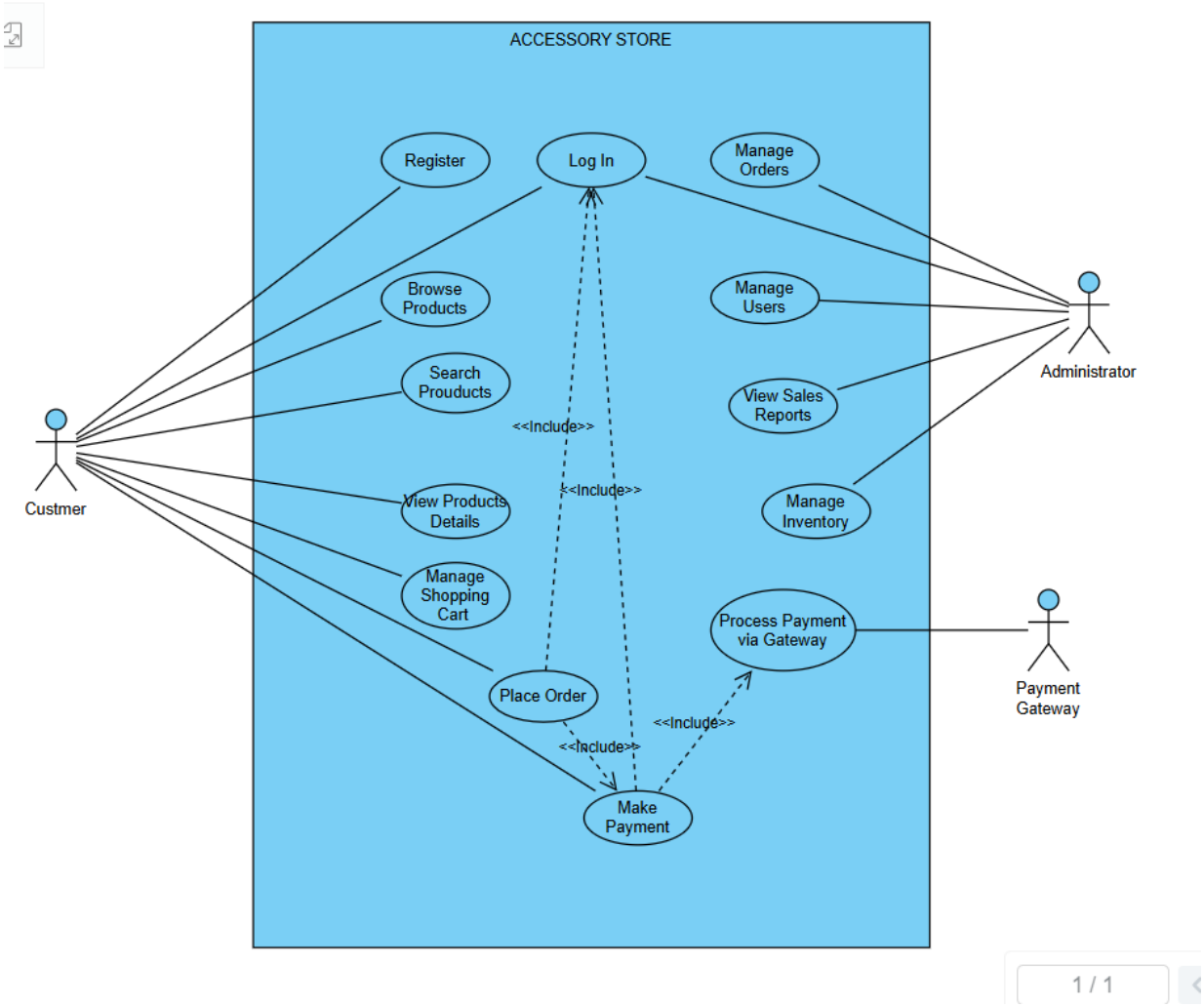
4.1 Identification of Actors

Actors are entities that interact with the Accessory Store system.

- **Primary Actors:**
  - **Customer (User):** An individual who interacts with the system to browse products, view details, register, log in, place orders, and make payments.
  - **Administrator (Store):** An employee or owner who manages the system's backend. Their responsibilities include updating inventory, managing user accounts, processing orders, tracking sales, and generating reports.
- **External Systems:**
  - **Payment Gateway:** An external system responsible for securely processing online payments.

4.2 Use Case Diagram

The following Use Case Diagram illustrates the functionalities available to the Customer and Administrator actors and their relationships.



4.3 Detailed Use Case Descriptions

Use Case 1: Register

Field	Description
Use Case Name	Register (UC1)
Actor(s)	Customer
Preconditions	The user is a guest (not logged in) and does not have an existing account.
Postcondition	A new Customer account is created and stored in the system. The user is authenticated and logged in.
Main Flow of Events	1. The Customer selects the "Register" option. 2. The system displays the registration form. 3. The Customer enters their required information (e.g., name, email, password). 4. The Customer submits the form. 5. The system validates the input (e.g., checks for unique email, password strength). 6. The system creates a new Customer record in database 7. The system creates a new session and logs the Customer in. 8. The system displays a registration message.

Use Case 2: Log In

Field	Description
Use Case Name	Log In (UC2)
Actor(s)	Customer, Administrator
Preconditions	The user has an existing account in the system and is currently logged out.
Postcondition	The system authenticates the user, creates a session, and grants them access to their respective functionalities (e.g., customer dashboard or admin panel).
Main Flow of Events	1. The user selects the "Log In" option. 2. The system displays the login form. 3. The user enters their credentials (email/username and password). 4. The user submits the form. 5. The system validates the credentials against the user database. 6. If valid, the system creates a user session. 7. The system redirects the user to their appropriate dashboard or homepage.

Use Case 3: Browse Products

Field	Description
Use Case Name	Browse Products (UC3)
Actor(s)	Customer
Preconditions	The Customer is on the website (can be a guest or logged in).
Postcondition	A list of products (e.g., in a specific category or all products) is displayed to the Customer.
Main Flow of Events	1. The Customer navigates to a product listing page (e.g., "Shop" or a specific category). 2. The system queries the database to retrieve the list of available products. 3. The system displays the products as a list or grid, showing key information (e.g., image, name, price).

Use Case 4: Search Products

Field	Description
Use Case Name	Search Products (UC4)
Actor(s)	Customer
Preconditions	The Customer is on the website.
Postcondition	A list of products matching the Customer's search query is displayed.
Main Flow of Events	1. The Customer enters a search term into the search bar. 2. The Customer submits the search query. 3. The system queries the Product database for items matching the search term. 4. The system displays the matching products on a search results page.

Use Case 5: View Product Details

Field	Description
Use Case Name	View Product Details (UC5)
Actor(s)	Customer
Preconditions	The Customer is viewing a list of products (e.g., from browsing or searching).
Postcondition	The system displays the full information for a single selected product.
Main Flow of Events	1. The Customer clicks on a product from a product list. 2. The system uses the product's ID to retrieve its full details from the database. 3. The system displays the product details page, showing information like description, price, larger images, and stock status.

Use Case 6: Place Order (UC6)

Field	Description
Use Case Name	Place Order (UC6)
Actor(s)	Customer
Preconditions	1. Customers are logged in. 2. The Customer has added at least one product to the cart.
Postcondition	An Order is created and marked as “Pending Payment”
Main Flow of Events	1. The Customer opens the shopping cart. 2. The system displays all items in the cart with quantities and prices. 3. The Customer reviews the items. 4. The Customer selects "Proceed to Checkout. 5. The system requests confirmation of shipping information (address, phone number). 6. The Customer confirms the order details. 7. The system creates a new Order and stores it with status “Pending Payment”. 8. The system redirects the Customer to the payment page.

Use Case 7: Make Payment (UC7)

Field	Description
Use Case Name	Make Payment (UC7)
Actor(s)	Customer, Payment Gateway (External System)
Preconditions	1. The Customer is logged in. 2. The Customer has proceeded to checkout and confirmed the items in their order.
Postcondition	The payment is processed successfully, the transaction is recorded, and the order is marked as paid.
Main Flow of Events	1. The system displays the payment form. 2. The Customer enters their payment information (e.g., credit card details). 3. The Customer confirms the payment. 4. The system securely sends the payment details and transaction amount to the external Payment Gateway.
	5. The Payment Gateway processes the transaction and returns a successful response. 6. The system receives confirmation and records the payment as successful. 7. The system updates the Order status to "Paid." 8. The system displays a payment success message to the Customer.

Use Case 8: Manage Inventory (UC8):

Field	Description
Use Case Name	Manage Inventory (UC8)
Actor(s)	Administrator
Preconditions	Administrator is logged in
Postcondition	Product list is updated (added, edited, or removed)
Main Flow of Events	1. The Administrator selects “Manage Inventory”. 2. The system displays the list of all products. 3. The Administrator selects one of the following actions: a. Add New Product b. Edit Existing Product c. Remove Product 4. The Administrator submits the changes. 5. The system validates the input (e.g., price cannot be negative). 6. The system updates the database accordingly. 7. A success message is displayed.

Use Case 9: Generate Sales Reports (UC9)

Field	Description
Use Case Name	Generate Sales Reports (UC9)
Actor(s)	Administrator
Preconditions	Administrator is logged in
Postcondition	A sales report is displayed or exported.
Main Flow of Events	1. The Administrator selects “Generate Sales Report”. 2. The system displays report filter options (date range, product category, etc.). 3. The Administrator selects the desired filters. 4. The system retrieves sales data from the database. 5. The system generates a report (summary + statistics). 6. The report is displayed and can be downloaded.

Use Case 10: Manage Users (UC10)

Field	Description
Use Case Name	Manage Users
Actor(s)	Administrator
Preconditions	Administrator is logged in User records exist in the database.
Postcondition	User accounts are updated, created, suspended, or deleted depending on the admin's actions.
Main Flow of Events	1. Admin opens the User Management panel. 2. System displays a list of all users (name, email, status, role). 3. Admin selects one of the available actions: 4. View user details 5. Edit user information 6. Suspend user 7. Delete user 8. Admin performs the desired operation. 9. System saves the changes and updates the user list.

Use Case 11: Manage Shopping Cart (UC11)

Field	Description
Use Case Name	Manage Shopping Cart
Actor(s)	Customer
Preconditions	User has access to the shopping cart page. Products exist in the database
Postcondition	Cart is updated based on the customer's actions (add, remove, update quantity, clear).
Main Flow of Events	1. User opens the Shopping Cart page. 2. System displays all items in the cart (product image, name, price, quantity, subtotal). 3. User performs one or more actions: • Update item quantity • Remove an item • Clear entire cart • Continue shopping • Proceed to checkout 4. System updates the cart accordingly. 5. System recalculates subtotal, tax, delivery fee (if any), and total amount. 6. Updated cart is displayed to the user.

Use Case 12: Manage Orders (UC12)

Field	Description
Use Case Name	Manage Orders
Actor(s)	Administrator
Preconditions	Admin is authenticated. Orders exist in the system.
Postcondition	Order details are updated (status, items, notes), or orders are canceled/refunded if required.
Main Flow of Events	1. Admin accesses the Order Management dashboard. 2. System shows all orders (order ID, customer, date, total, status). 3. Admin selects a specific order. 4. System displays order details (items, quantities, payment status, shipping info). 5. Admin updates one or more of the following: • Order status (Pending → Processing → Shipped → Delivered → Completed) • Cancel the order • Refund the order • Update shipping details • Add internal admin notes

4.4 Identification of Classes

Based on the system requirements, the following core classes are identified:

- **User:** A base class that manages common data for all system users, such as login credentials.
- **Customer:** A specialized type of User who can browse products, view details, and place orders.
- **Administrator:** A specialized type of User responsible for managing the system, including updating inventory, processing orders, and managing user accounts.
- **Product:** Represents an accessory item available for sale. It stores information like price and details.
- **Order:** Represents a customer's request to purchase one or more products. It tracks the status of the order.
- **OrderItem:** An association class that links a specific Product to an Order and stores the quantity of that product in the order.
- **Payment:** A class responsible for processing the payment for an Order and interacting with the external payment gateway.

4.5 Relationships

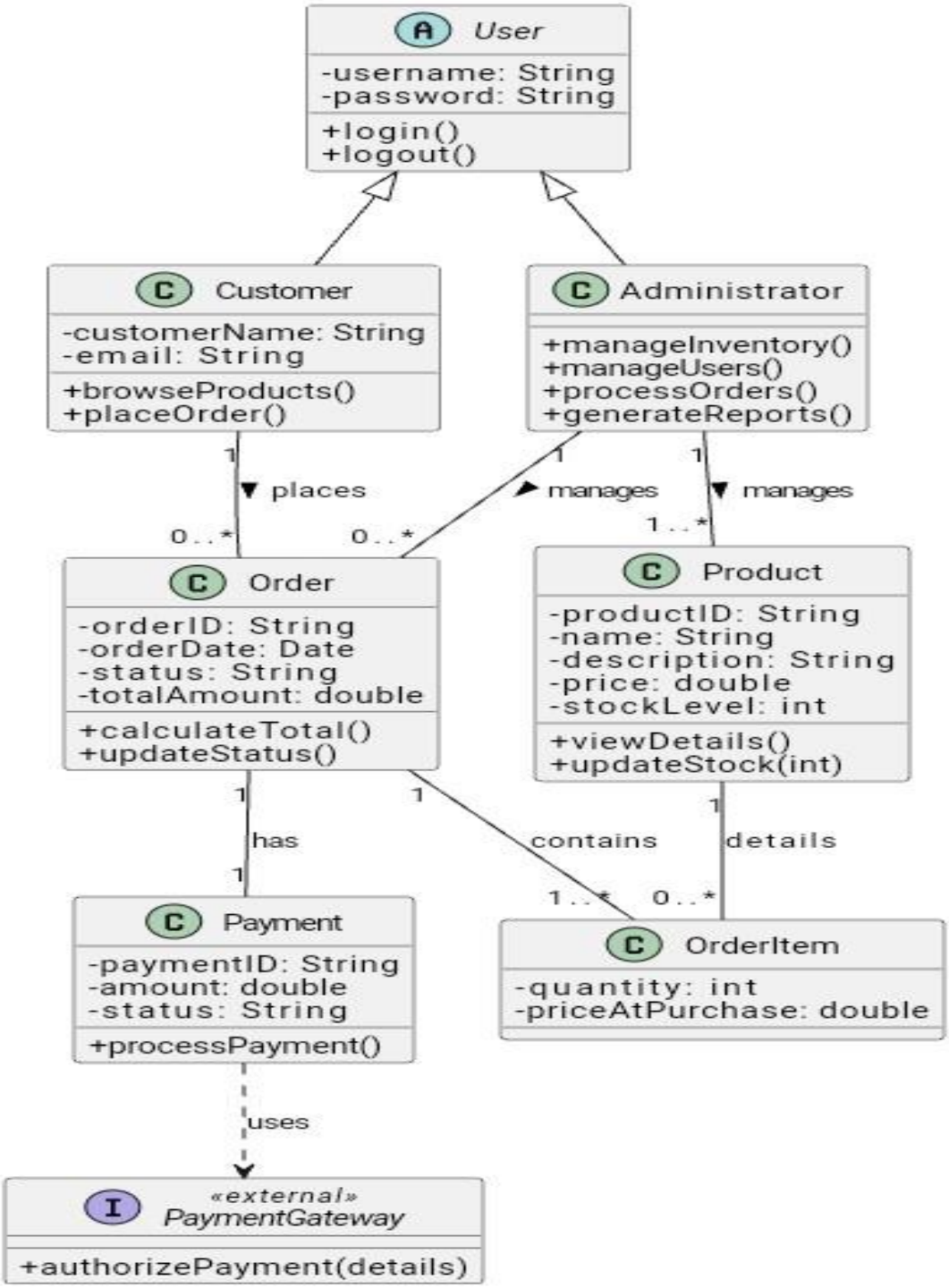
The classes identified above are related in the following ways:

- **Generalization/Inheritance:**
  - Customer **is a** User.
  - Administrator **is a** User.
- **Association:**
  - A Customer **places** zero or more Orders (0..*).
  - An Order **is placed by** exactly one Customer (1).
  - An Administrator **manages** one or more Products (1..*).
  - An Administrator **manages** zero or more Orders (0..*).
  - An Order **contains** one or more OrderItems (1..*).
  - A Product **is part of** zero or more OrderItems
  - An Order **has** exactly one Payment (1).
- **Dependency:**

The Payment class **uses** (depends on) the external Payment Gateway to process transactions.

4.6 Class Diagram

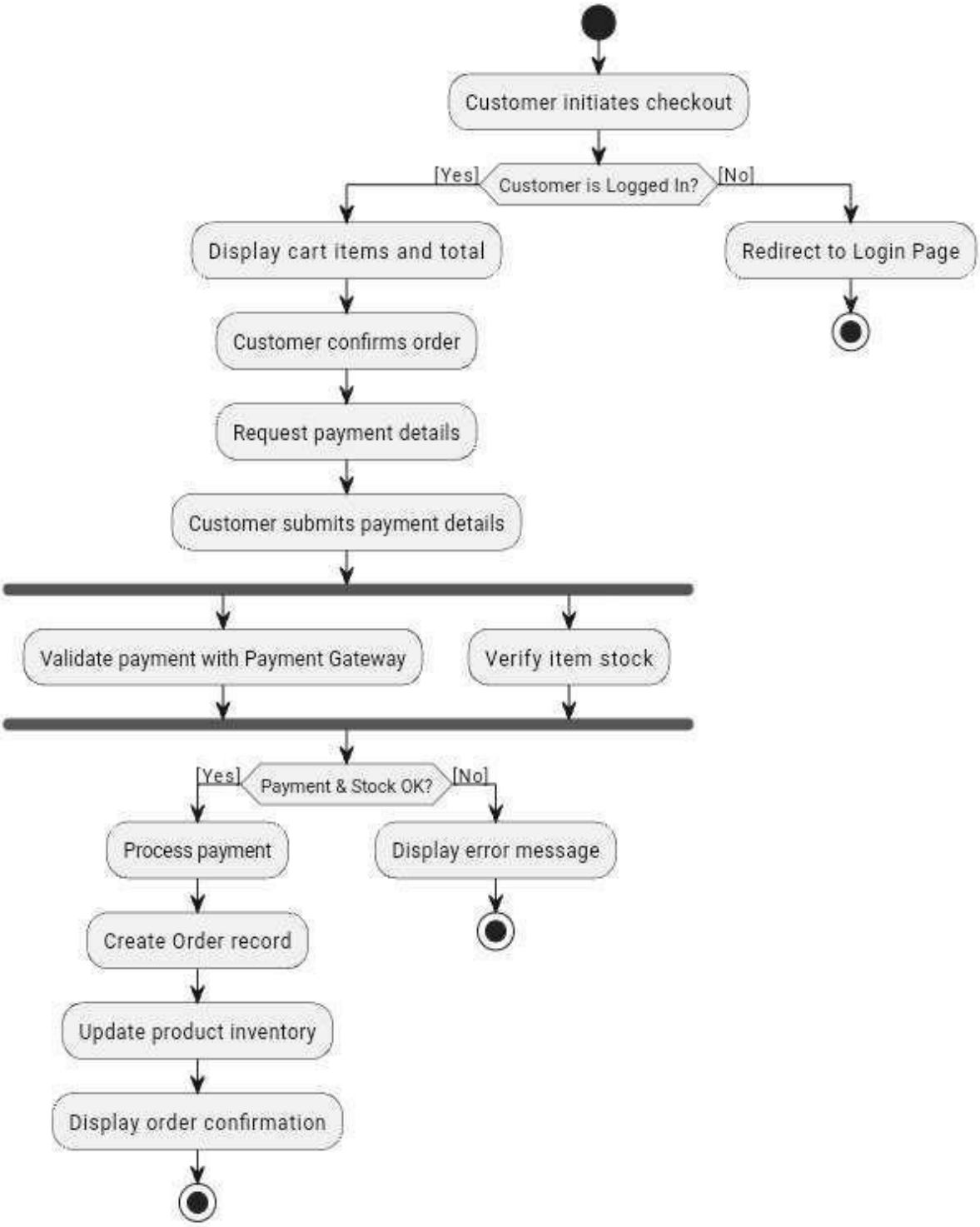
The class diagram illustrates the main classes and their relationships within the **Accessory Store System**. It shows how different components—such as customers, orders, products, and employees—interact with each other.



CHAPTER 5: OBJECT-ORIENTED MODELLING

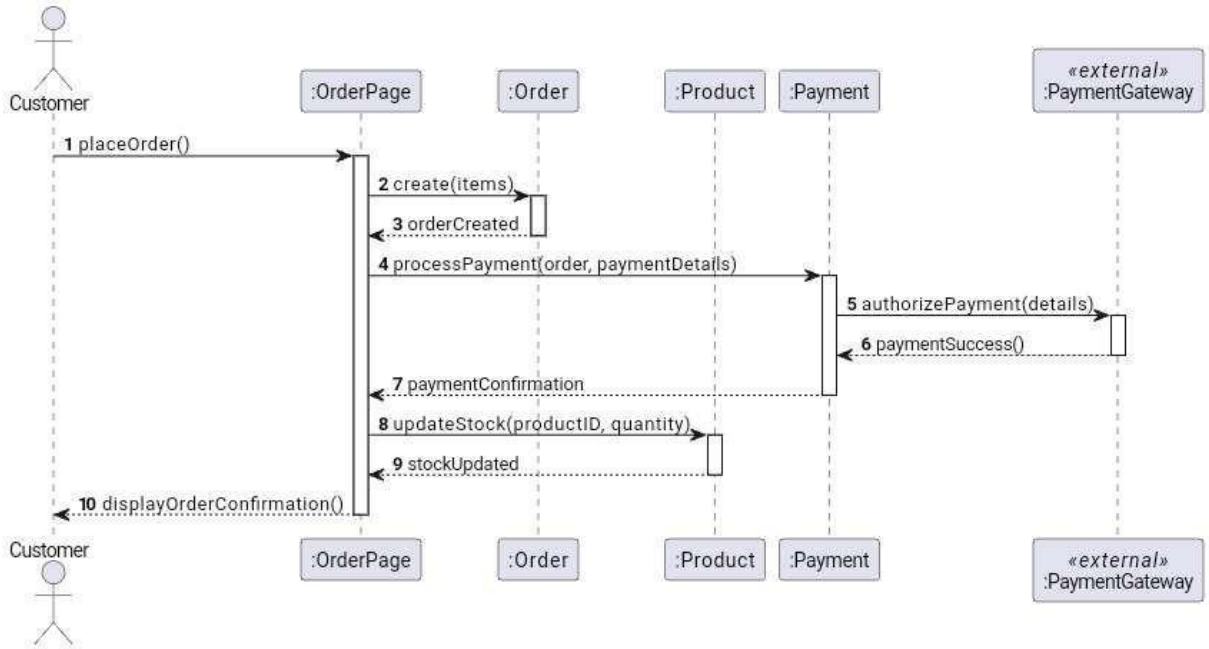
5.1 Activity Diagram

The following activity diagram illustrates the workflow for the **"Place Order"** use case.



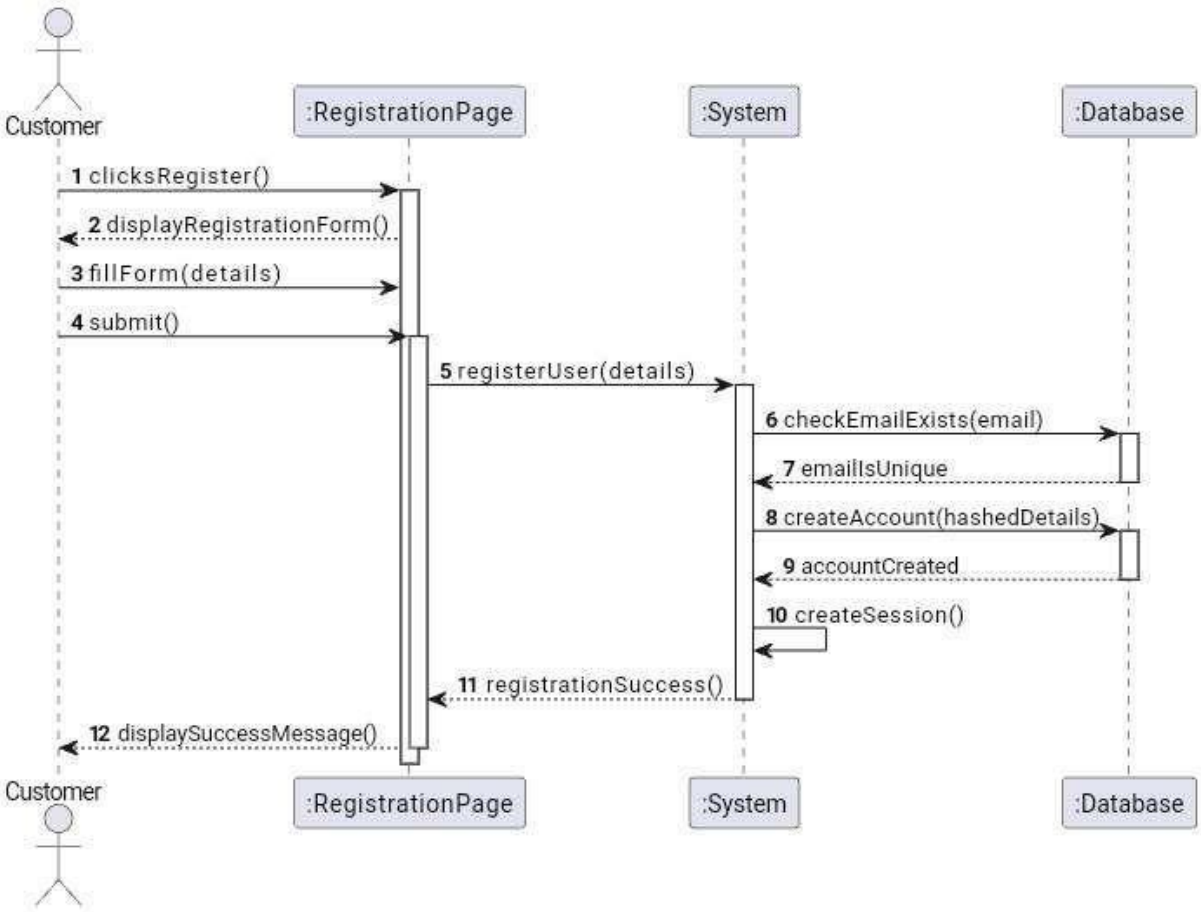
5.2 Sequence Diagram

The following sequence diagram shows the interactions between objects over time for the **"Place Order"** use case.

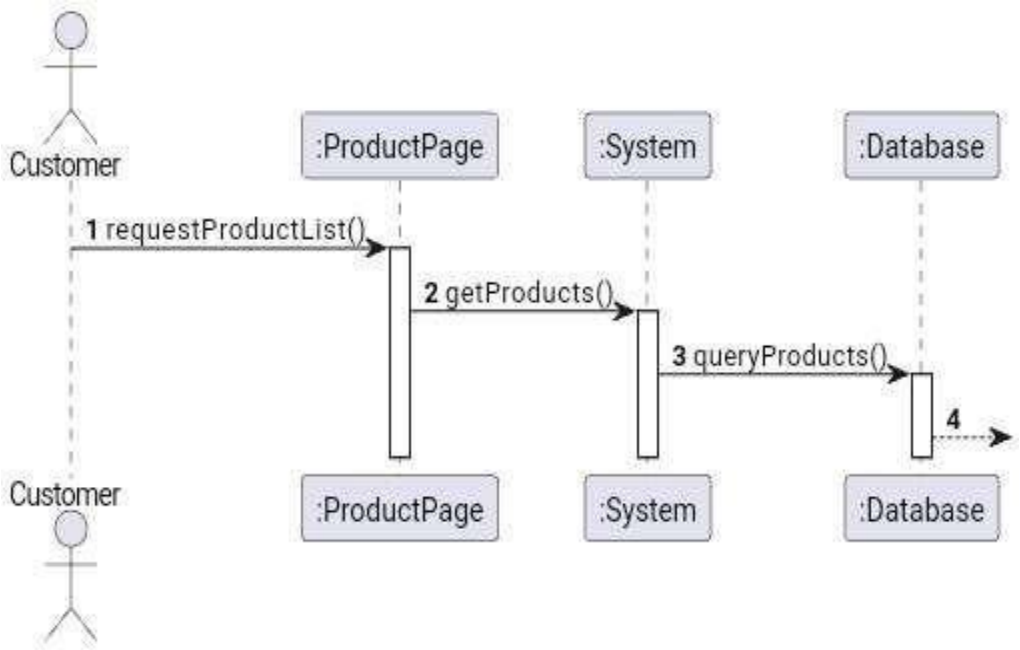


Sequence Diagram: Register (UC1)

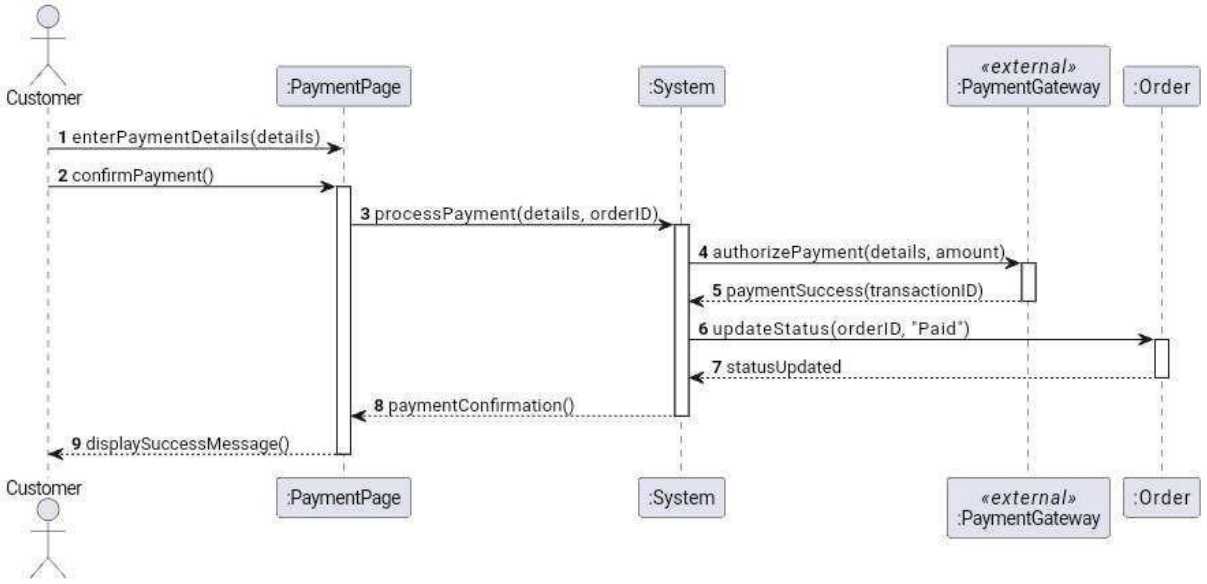
1.



2. Sequence Diagram: Browse Products (UC3)

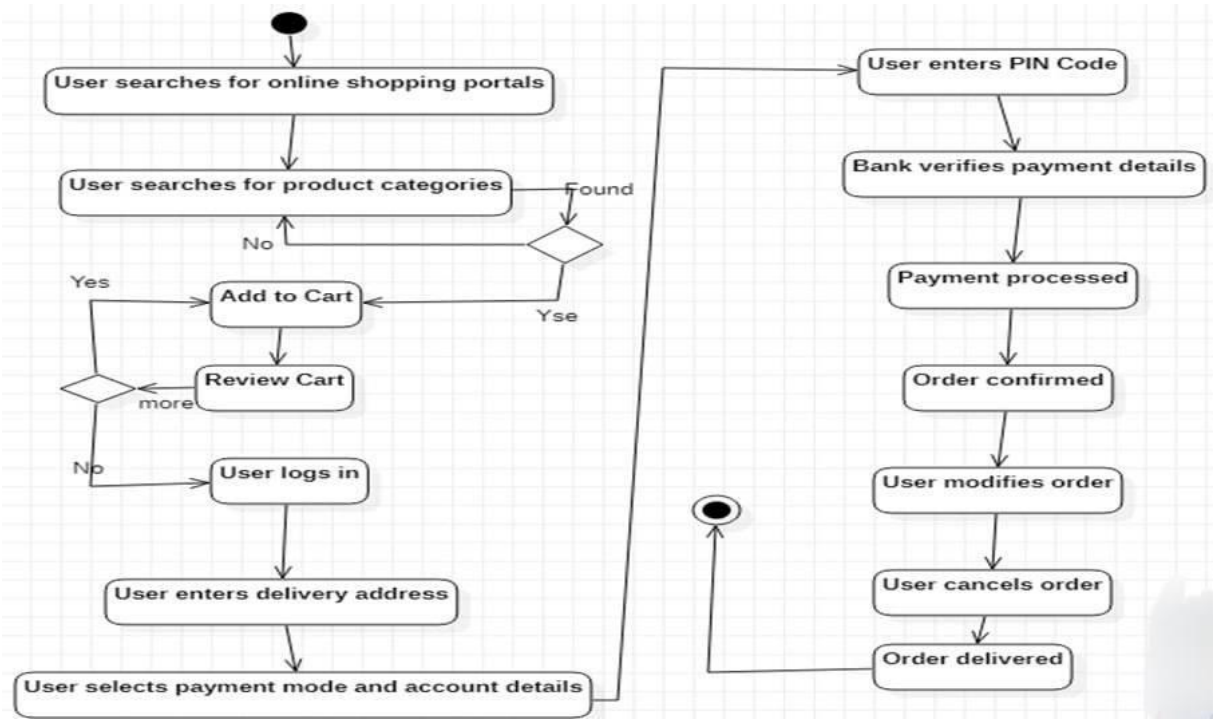


3. Sequence Diagram: Make Payment (UC7)



5.3 State Diagram

The state diagram illustrates the complete online shopping workflow in the Accessory Store Project. It begins with the user searching for products on the platform, selecting items, and adding them to the cart. The user can continue browsing or proceed to review the cart, after which they log in and enter their delivery address. The system then moves to the payment stage, where the user provides payment details and a PIN code. The bank verifies the information, processes the payment, and confirms the order. After confirmation, the user may modify or cancel the order. Finally, the process ends with the successful delivery of the order. This diagram shows the sequential and decision-based transitions that guide the user from browsing accessories to completing a secure purchase and receiving their product.



CHAPTER 6: CONCLUSION

Summary of What Was Achieved

This project successfully applied software engineering principles to the design and documentation of an "Accessory Store" e-commerce platform. The primary goal was to create a system that addresses the challenges small accessory shops face in reaching a wider audience and provides customers with an efficient way to find and purchase high-quality products.

The project followed the Software Development Life Cycle (SDLC), resulting in a comprehensive set of deliverables. This includes a clear problem statement, defined objectives, and a detailed scope. A project plan with a timeline and task allocation was established. The system's functional and non-functional requirements were specified, followed by a complete object-oriented design, including Use Case, Class, Activity, and Sequence diagrams. The project focused strictly on the design and documentation phases, as per the project constraints.

Key Learning Outcomes

This project served as a significant training opportunity to apply theoretical software engineering concepts in a practical, real-world context. Key skills developed by the team include:

- **System Analysis and Design:** Translating business needs (e.g., efficient inventory management, user registration) into formal models.
- **UML Modeling:** Gaining proficiency in creating diagrams to represent system structure and behavior.
- **Documentation:** Understanding the importance of clear, consistent, and thorough documentation for a software project.
- **Teamwork and Project Management:** Collaborating as a team to manage tasks, meet milestones, and deliver a cohesive project.
- **Constraint Awareness:** Working within defined project limitations, such as time and the constraint of delivering only design and documentation.

Possible Future Improvements/Extensions

The current document provides a solid foundation for a prototype. Several improvements could be made in a future implementation phase:

- **Full Implementation:** The most obvious extension is the full development, testing, and deployment of the system, which was outside the current project's scope.
- **Enhanced User Interface:** The system could be expanded to include a more enhanced and interactive user interface.
- **Multi-language Support:** To reach a wider international audience, the platform could be developed to support multiple languages.
- **Advanced Analytics:** Implementing an advanced sales and user behavior analytics dashboard for administrators, beyond the basic sales reports.
- **Mobile Application:** Developing native mobile applications (iOS/Android) to complement the web platform and improve customer accessibility.

REFERENCES

[1] Fowler, M. (2004). *UML Distilled: A Brief Guide to the Standard Object Modeling Language* (3rd ed.). Addison-Wesley Professional.

[2] Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.

[3] Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.

[4] The Object Management Group. (2017). *OMG Unified Modeling Language (OMG*